Actual quotient functional and coefficient pencil

November 13, 2026. Unsigned producer evidence [pcc:unverified].

New assembly component. Over any field let F, H be monic, $\deg H > 0$, $\deg H + 1 < \deg F$, and suppose

$$UF + V(fH) = 1, \quad (\alpha F)^2 - f(\beta H)^2 = cG, \quad c \neq 0.$$

Define the actual linear functional $\mu(P) = [X^{\deg F - 1}]((PV) \bmod F)$. Lean derives $\mu(G) = 0$ from the supplied norm equation. It is not an assumed pencil-membership hypothesis.

For monic G, $d = \operatorname{natDegree} G$, and trial A, B with

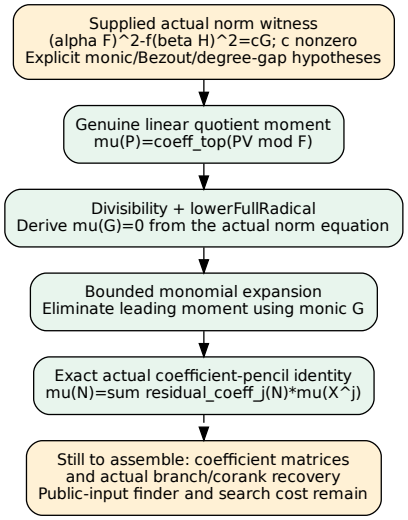
$$2 \operatorname{natDegree} A \leq d, \qquad \operatorname{natDegree} f + 2 \operatorname{natDegree} B \leq d,$$

let $N = A^2 - fB^2$. The exact functional coefficient-pencil identity is

$$\mu(N) = \sum_{j<d} \big([X^j]N - [X^d]N\,[X^j]G\big)\mu(X^j).$$

This supplies the actual leading-eliminated coefficient constraints. It does not yet identify a finite matrix or compute the moments from public inputs: the construction still uses supplied witness polynomials.

Proof. Divisibility gives $\mu((\alpha F)^2) = 0$. The preceding full lower radical, tested on H, gives $\mu(fH^2) = 0$; linearity scales it by β^2. Applying μ to the norm equation and cancelling c gives $\mu(G) = 0$. Expand N and monic G through degree d and eliminate the leading moment. The existing actual norm-degree theorem supplies the trial bound. No characteristic or separability condition is added.



Proof map, checks and remaining dependencies

Source. `NormPencilFunctional.lean`, frozen source commit `78123b01d`. Definitions and declarations: `moment`, `momentApply`, `annihilatesTarget`, `coefficientExpansion`, `coefficientPencil`, `normCoefficientPencil`, `normPolarization`. Five substantive interpretive PCC records are unsigned; exact declaration hashes and scope are retained. Earlier CRT/radical/moment hashes are unchanged.

Checks. Lean4.35.0, unchanged pinned CryptoLean and Mathlib. Build004 passes3147 jobs. Final tracked-source gate covers110 declarations, zero failing and zero additional trusted axioms. Fresh kernel replay including imports exits0. Standard foundations only: `propext`, `Classical.choice`, `Quot.sound`. No signed PCC standing is claimed.

Build001 failed on exported definitional-equality visibility of the new functional; an explicit simp proof resolves it. Build003 failed the missing import of the reused degree theorem; its actual module is now imported. Build002 was successful at the pre-specialization scope. All logs, including failed attempts, are preserved in the evidence directory.

Native identity. The universal commutative-ring polarization is

$$\mathrm{Norm}(A+Q,B+T)-\mathrm{Norm}(A,B)-\mathrm{Norm}(Q,T)=2(AQ-fBT).$$

Native record `IDC1h345f3824264d232a`, with32 statement-derived and32 extra-stream points, pins this exact universal Lean source. The maintained issuer/checker is unchanged. Its adapter's22 inherited tests pass, including tamper/statement-swapping controls. It does not execute Lean or establish the conditional coefficient-pencil theorem. Division by2 is not used here.

Next. Define actual bounded coefficient bases and derive matrix pencil equality from the functional identity. Instantiate the branch lemma, compute the scaled witness-line intersection, then transfer its dimension to corank1 and guarded adjugate recovery. Constant/degree-drop charts and a public-input finder remain separate. The required $2^{o(n)}$ bit search cost, $n=\lceil \log_2 r \rceil$, does not follow from certificate existence.

Earlier completion proposal. The nonscalar origin/jet-completion idea motivated this rank-directed certificate design. These identities do not prove reachability through $W_{\mathrm{base}}+C(T)H(T)$ or provide an efficient choice of its free coefficients.

Sources and resources. Previous matrix-bridge plan and exact pinned `Polynomial.RingDivision` scalar-remainder/linear-map declarations, `Degree.Support` bounded monomial expansion, prior degree bound and lower radical. KB and conductor unavailable; bounded fallback recorded, no priority claim for these classical ingredients. No paid job, empirical runtime/scaling campaign or budget override. Monthly isolab budget exhausted. No benchmark, leaderboard or ecdb changes. The source-linked Markdown report, editable DOT/SVG, diagram PDF and evidence accompany this PDF.